

Gini Impurity in Decision Trees

Gini impurity measures how “mixed” the classes are in a node.

Decision trees split data by finding the feature and threshold that best separates classes.

Goal: Choose splits that minimize Gini impurity, creating purer child nodes.

Gini Impurity Formula

General formula for C classes:

$$\text{Gini}(D) = 1 - \sum_{i=1}^C p_i^2$$

Binary classification ($C = 2$):

$$\text{Gini}(D) = 1 - p_0^2 - p_1^2$$

where p_0 and p_1 are the proportions of class 0 and class 1 in node D .

Dataset

Sample	Feature A	Feature B	Label
1	0	1	0
2	0	0	0
3	0	1	0
4	1	0	0
5	1	1	1
6	1	0	1
7	1	1	1
8	1	1	1

Label distribution: 4 samples with Label=0, 4 samples with Label=1

Parent Node Gini Impurity

Class proportions:

$$p_0 = \frac{4}{8} = 0.5, \quad p_1 = \frac{4}{8} = 0.5$$

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Interpretation: Maximum impurity (50/50 split between classes)

Split on Feature A

Left child (Feature A = 0): Samples 1, 2, 3 $\rightarrow$ Labels: 0, 0, 0

$$\text{Gini}(\text{left}) = 1 - 1^2 - 0^2 = 0 \quad (\text{pure node})$$

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Right child (Feature A = 1): Samples 4, 5, 6, 7, 8 $\rightarrow$ Labels: 0, 1, 1, 1, 1

$$p_0 = 0.2, \quad p_1 = 0.8$$

$$\text{Gini}(\text{right}) = 1 - 0.2^2 - 0.8^2 = 0.32$$

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$$\text{Gini}(\text{right}) = 1 - 0.2^2 - 0.8^2 = 0.32$$

Weighted Gini:

$$\text{Gini}_A = \frac{3}{8} \times 0 + \frac{5}{8} \times 0.32 = 0.2$$

Split on Feature B

Left child (Feature B = 0): Samples 2, 4, 6 $\rightarrow$ Labels: 0, 0, 1

$$\text{Gini}(\text{left}) = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{4}{9} \approx 0.444$$

Split on Feature B

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Right child (Feature B = 1): Samples 1, 3, 5, 7, 8 $\rightarrow$ Labels: 0, 0, 1, 1, 1

$$\text{Gini}(\text{right}) = 1 - 0.4^2 - 0.6^2 = 0.48$$

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Right child (Feature B = 1): Samples 1, 3, 5, 7, 8 $\rightarrow$ Labels: 0, 0, 1, 1, 1

$$\text{Gini}(\text{right}) = 1 - 0.4^2 - 0.6^2 = 0.48$$

Weighted Gini:

$$\text{Gini}_B = \frac{3}{8} \times 0.444 + \frac{5}{8} \times 0.48 = 0.467$$

Comparison & Decision

Split	Weighted Gini
Feature A	0.200
Feature B	0.467

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Decision: Split on Feature A (lower Gini impurity)

Feature A perfectly separates all Label=0 samples, while Feature B mixes classes in both children.

Key Takeaways

- **Gini = 0** → Pure node (all samples belong to one class)
- **Gini = 0.5** → Maximum impurity for binary classification
- Decision trees select the split that **minimizes weighted Gini impurity**
- Weighted Gini accounts for child node sizes